

LumanSense Edge AI - Zone Transition Calculations Report

1. Raw Telemetry Input Data

Below is the raw pedestrian vehicle counts per zone used as input for the calculations. The junctions are mapped as: Junction 1 -> Zone A, Junction 2 -> Zone B, Junction 3 -> Zone C, Junction 4 -> Zone D (omitted/0 in this sample).

DateTime	Zone A (Junc 1)	Zone B (Junc 2)	Zone C (Junc 3)	Zone D (Junc 4)
00:00:00 (t0)	15	6	9	0
01:00:00 (t1)	13	6	7	0
02:00:00 (t2)	10	5	5	0

2. Hourly Interval-by-Interval Transitions (Independent Steps)

Calculated independently for each hour transition interval using: $\text{Transition}(i \rightarrow j) = \min(\text{Vehicles_t}[i], \text{Vehicles_t+1}[j])$, then normalized over total outgoing transitions for that zone.

Interval	A -> A	A -> B	A -> C	B -> A	B -> B	B -> C	C -> A	C -> B	C -> C
00:00 -> 01:00 (Interval 1)	$\min(15,13)=13$	$\min(15,6)=6$	$\min(15,7)=7$	$\min(6,13)=6$	$\min(6,6)=6$	$\min(6,7)=6$	$\min(9,13)=9$	$\min(9,6)=6$	$\min(9,7)=7$
	$13/26 =$	$6/26 =$	$7/26 =$	$6/18 =$	$6/18 =$	$6/18 =$	$9/22 =$	$6/22 =$	$7/22 =$
	50.00%	23.08%	26.92%	33.33%	33.33%	33.33%	40.91%	27.27%	31.82%
01:00 -> 02:00 (Interval 2)	$\min(13,10)=10$	$\min(13,5)=5$	$\min(13,5)=5$	$\min(6,10)=6$	$\min(6,5)=5$	$\min(6,5)=5$	$\min(7,10)=7$	$\min(7,5)=5$	$\min(7,5)=5$
	$10/20 =$	$5/20 =$	$5/20 =$	$6/16 =$	$5/16 =$	$5/16 =$	$7/17 =$	$5/17 =$	$5/17 =$
	50.00%	25.00%	25.00%	37.50%	31.25%	31.25%	41.18%	29.41%	29.41%

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3. Cumulative Transition Probabilities (Aggregated History)

Shows the rolling probabilities as new interval transitions are aggregated into history.

Cumulative Range	A -> A	A -> B	A -> C	B -> A	B -> B	B -> C	C -> A	C -> B	C -> C
After Interval 1 (00:00 -> 01:00)	13 / 26 = 50.00%	6 / 26 = 23.08%	7 / 26 = 26.92%	6 / 18 = 33.33%	6 / 18 = 33.33%	6 / 18 = 33.33%	9 / 22 = 40.91%	6 / 22 = 27.27%	7 / 22 = 31.82%
After Interval 2 (00:00 -> 02:00)	(13+10)/46 = 23/46 = 50.00%	(6+5)/46 = 11/46 = 23.91%	(7+5)/46 = 12/46 = 26.09%	(6+6)/34 = 12/34 = 35.29%	(6+5)/34 = 11/34 = 32.35%	(6+5)/34 = 11/34 = 32.35%	(9+7)/39 = 16/39 = 41.03%	(6+5)/39 = 11/39 = 28.21%	(7+5)/39 = 12/39 = 30.77%

4. Multi-step Vector-Matrix Multiplication (3 Hops from Zone A)

A state vector V_t represents occupancy probabilities for zones [A, B, C, D] at step t. At each step, the distribution is updated by multiplying the vector by the transition matrix P: V_t = V_t-1 * P. Below is the trace starting from Zone A (V_0 = [1.0, 0.0, 0.0, 0.0]) using the final cumulative matrix.

Step / Hop	Zone A Probability Calculation	Zone B Probability Calculation	Zone C Probability Calculation	Zone D	Result Vector (V_t)
Hop 1 (V_1)	(1.0 * 0.5000) = 0.5000	(1.0 * 0.2391) = 0.2391	(1.0 * 0.2609) = 0.2609	0.0000	[0.5000, 0.2391, 0.2609, 0.0]
Hop 2 (V_2)	(0.5000 * 0.5000) + (0.2391 * 0.3529) + (0.2609 * 0.4103) = 0.4414	(0.5000 * 0.2391) + (0.2391 * 0.3235) + (0.2609 * 0.2821) = 0.2705	(0.5000 * 0.2609) + (0.2391 * 0.3235) + (0.2609 * 0.3077) = 0.2881	0.0000	[0.4414, 0.2705, 0.2881, 0.0]
Hop 3 (V_3)	(0.4414 * 0.5000) + (0.2705 * 0.3529) + (0.2881 * 0.4103) = 0.4344	(0.4414 * 0.2391) + (0.2705 * 0.3235) + (0.2881 * 0.2821) = 0.2743	(0.4414 * 0.2609) + (0.2705 * 0.3235) + (0.2881 * 0.3077) = 0.2913	0.0000	[0.4344, 0.2743, 0.2913, 0.0]